

CHG 569

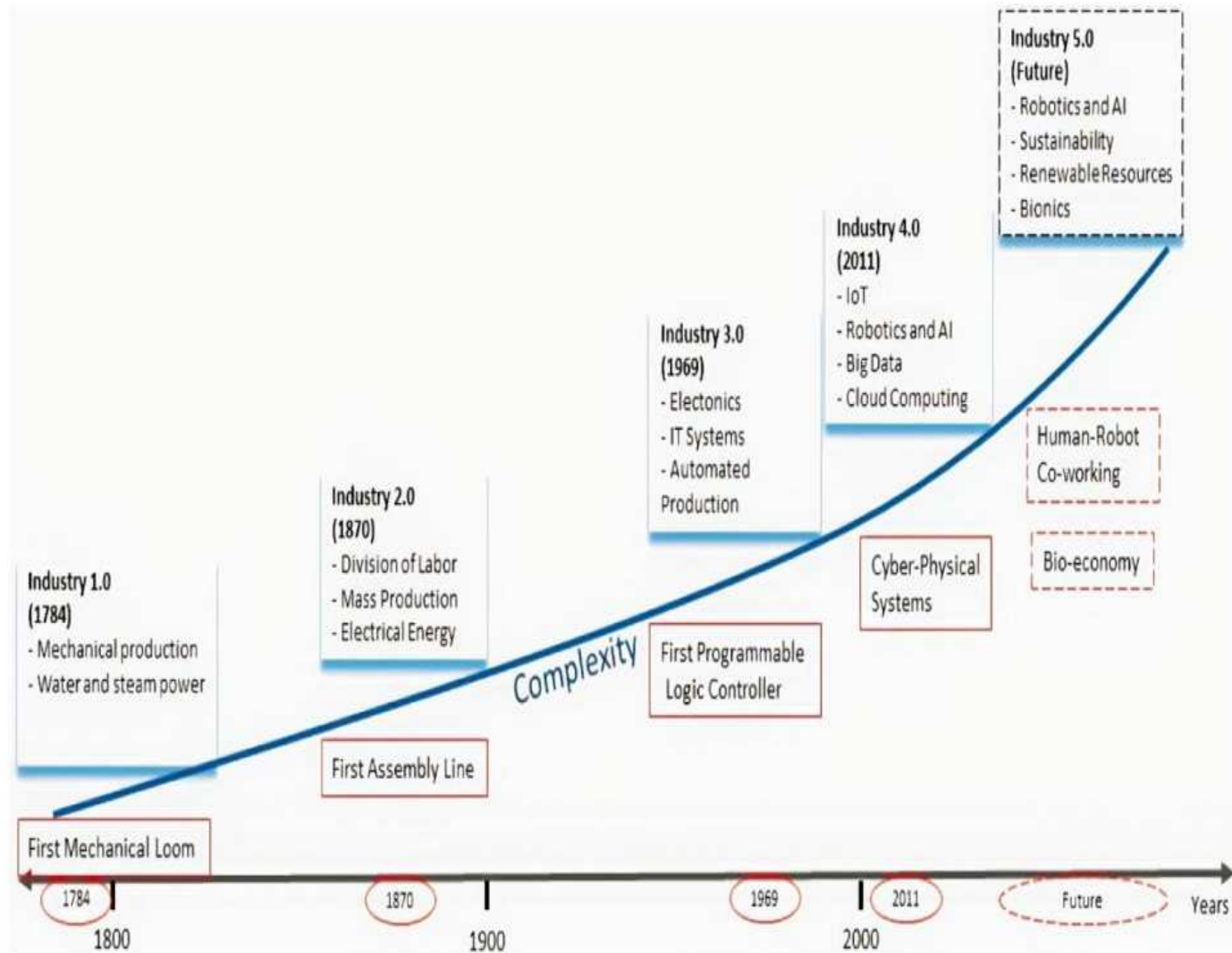
Environmental Pollution Engineering

Module XI
The way forward
for man and the environment

Biomimicry
for
Sustainable Development

The journey so far







4th

Cyber Physical
Systems



3rd

Computer and
automation



1st

Mechanization,
water power, steam
power



2nd

Mass production,
assembly line,
electricity

Industrial Revolution and Society 5.0

1.0 Mechanisation



- The introduction of industrial production equipment driven by water and steam power.

1780

2.0 Electrification



- Mass production using electrical energy and assembly lines.

1870

3.0 Automation



- Automated production due to the rise of electronics, telecommunications and computers.

1970

4.0 Digitalisation



- The use of cyber physical systems on connected devices to automate processes further.

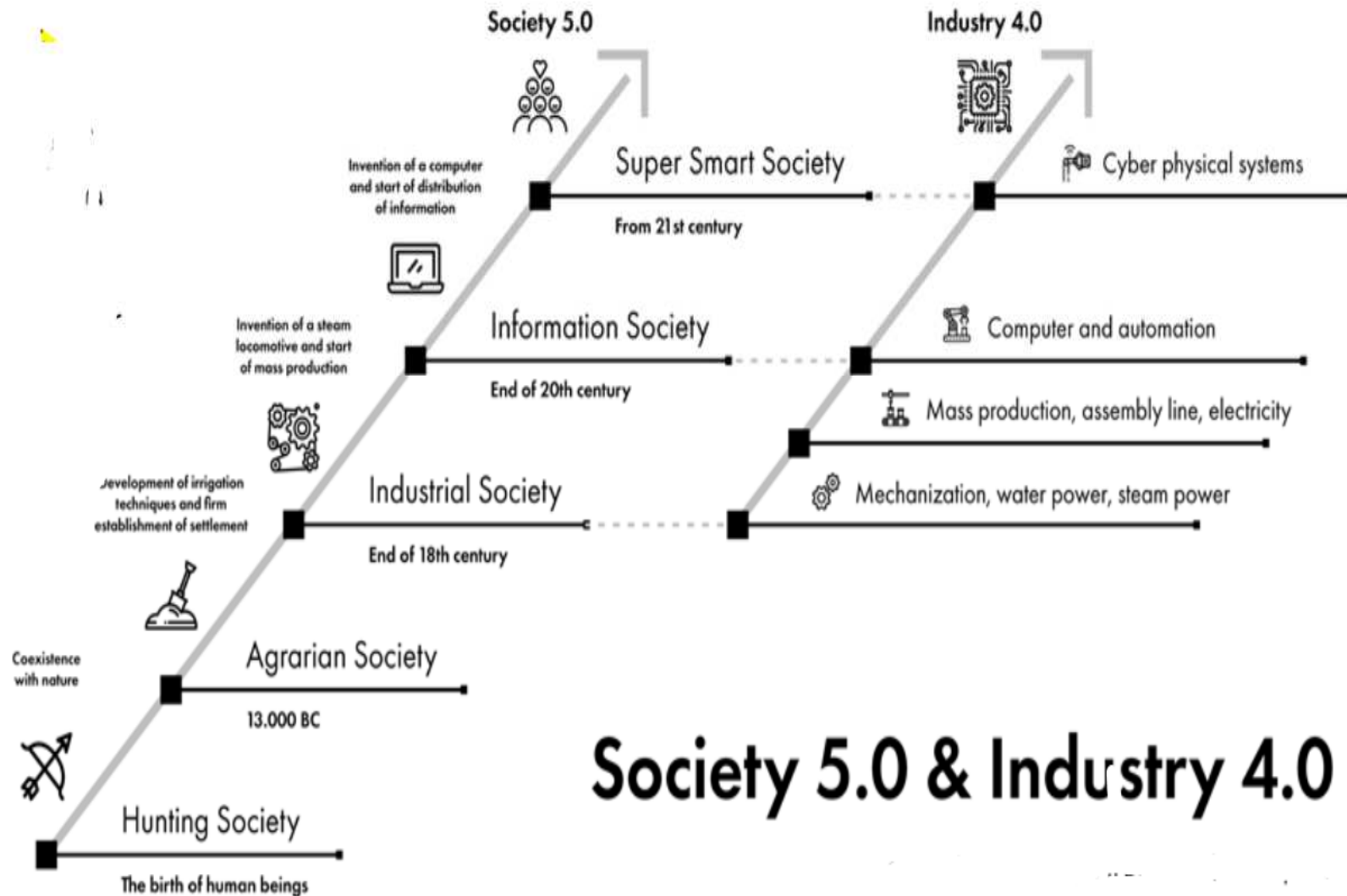
2011

5.0 Personalisation



- The interdependence of man and machine using cognitive computing and human intelligence. Mass customization and personalization for humans.

2020



Society 5.0 & Industry 4.0

Industry 4.0 and the environment

The negative impacts of digitalization are:

- **Digital obsolescence** — The rapid obsolescence of devices and equipment will result in **significant electronic waste**
- **Increased energy consumption** — The use of advanced technologies like AI, robotics, IoT devices lead to **higher energy demand**
- **Waste streams** - Industrial processes and transportation related to Industry 4.0 also cause **air, water and land pollution**
- **Resource depletion**— The extraction and processing of rare earth metals and other resources for advanced technologies can harm ecosystems.

The Problem

The revolutions that have occurred over time has changed man's relationship with nature:

- ▶ **Agricultural revolution** – from the hunter/gatherer to the mechanized farmer
- ▶ **Scientific revolution** – man could scientifically explain the workings of nature and use for innovations and design
- ▶ **Industrial revolution** – industries were as varied as they were as diverse, with their attendant environmental problems
- ▶ **Robotic revolution** – machines now perform man's task.

Science has helped man to explore, exploit and manipulate nature, but it has also failed man; man impacted nature negatively and nature is now distressed and affecting man negatively.

"The world we have made, as a result of the level of thinking we have done thus far, creates problems we cannot solve at the same level of thinking at which we created them."

– Albert Einstein.

The Solution

The activities man undertakes for survival on planet earth is also being undertaken by the plants and animals that he shares the earth with.

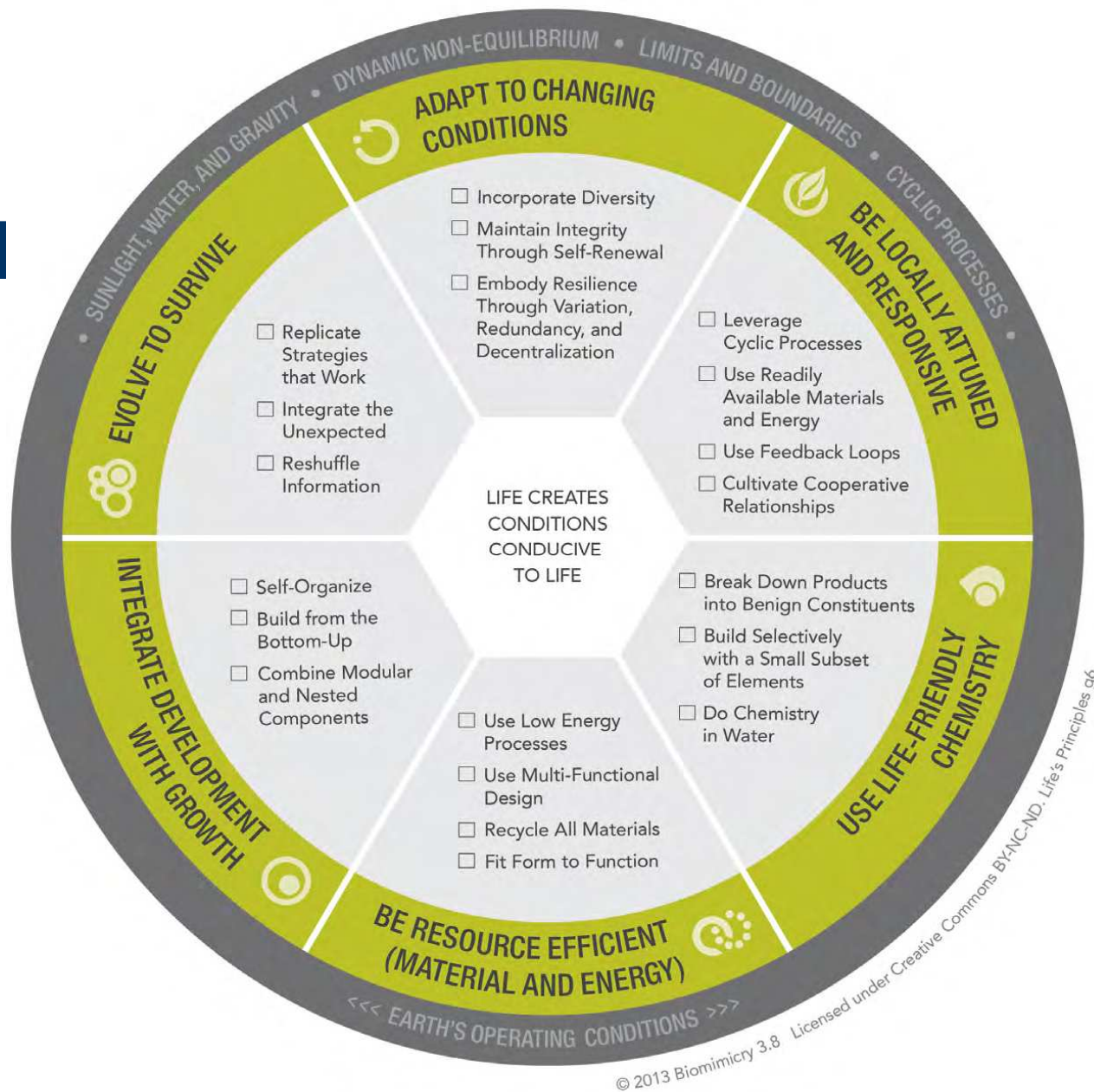
Nature has had 3.8 billion years of experience and man can draw inspiration for innovation from her, this is the paradigm shift needed by man to save the earth thus ensuring sustainability.

“Nature is the source of all true knowledge. She has her own logic, her own laws, she has no effect without cause nor invention without necessity.” - Leonardo DaVinci.

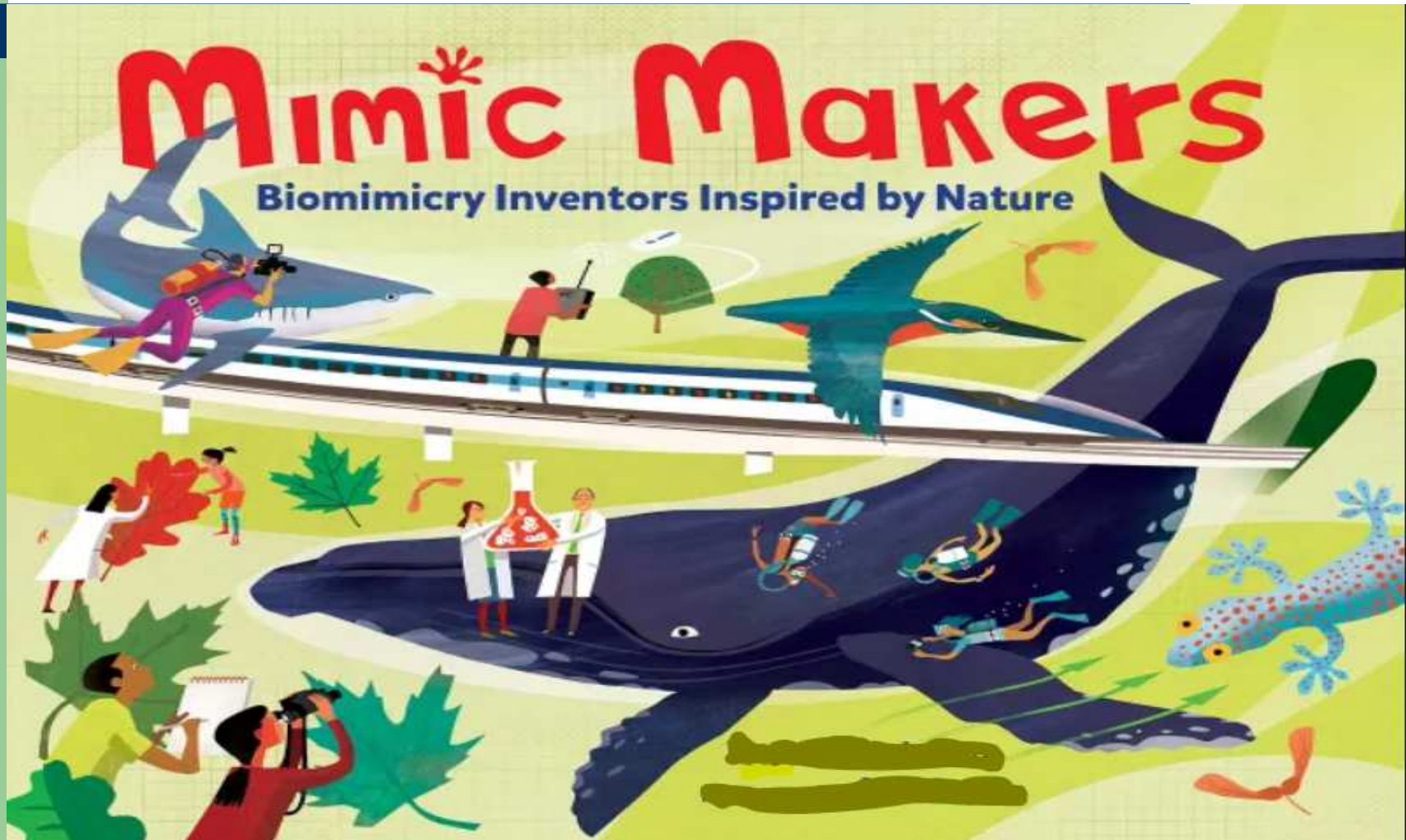
Nature's guiding principles

- Runs on sunlight
- uses only what is needed
- abhors waste and recycles everything
- form follows function
- collaborates and is multidisciplinary
- uses local expertise
- works within safe operating conditions
- runs on information.

Nature's inherent rule for sustainability



Biomimicry to the rescue



What is Biomimicry?

- Bio – life, nature
- Mimic - imitate

The practice of
observing and **imitating**
nature's designs and **processes**
to solve human challenges.

- Stealing ideas from nature
- Innovation inspired by nature

Biomimicry

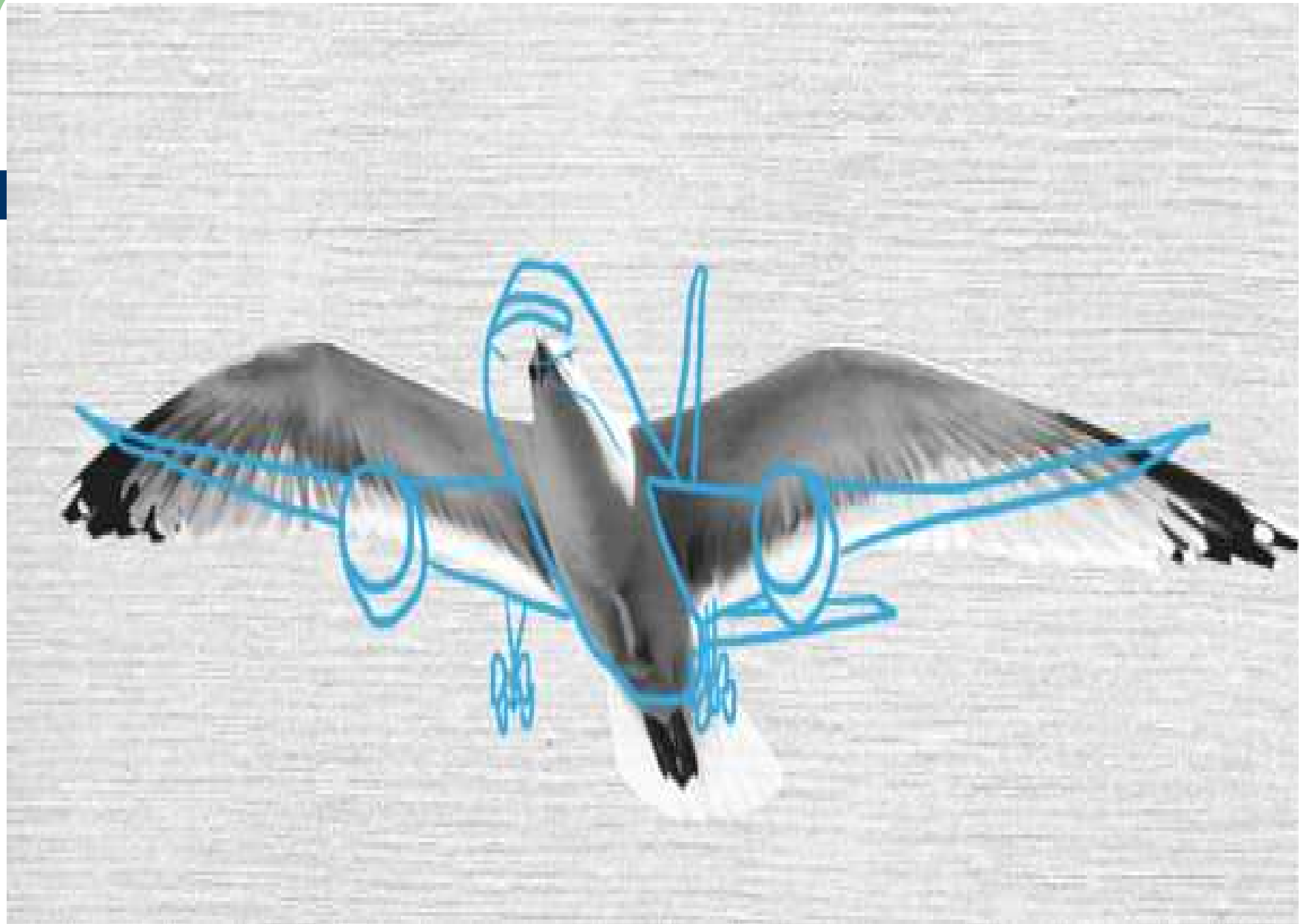
By imitating nature's efficient designs, processes, and systems, biomimicry helps reduce resource consumption in various industries. This approach promotes the use of sustainable materials and manufacturing processes, minimizing waste generation and conserving valuable resources.

The 3Ms of biomimicry

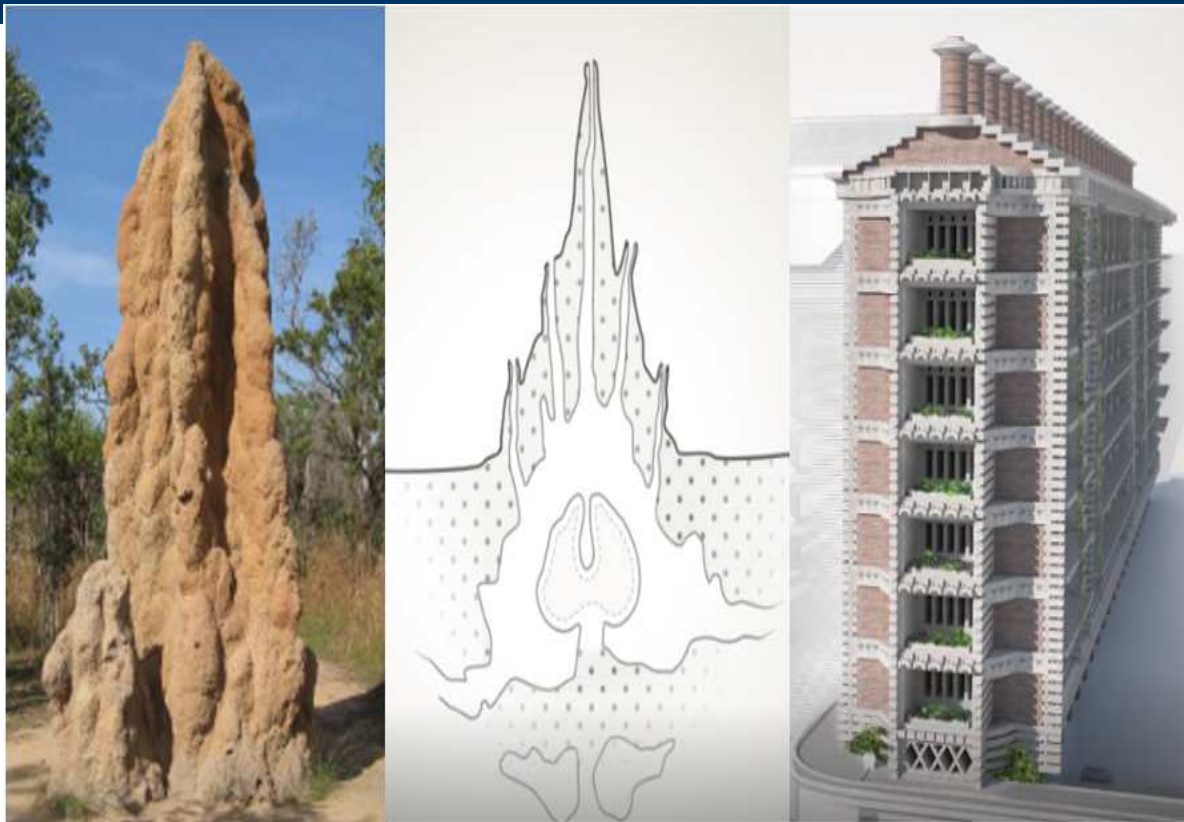
The process of inculcating these principles necessitates seeing nature as a **model**, a **measure**, and a **mentor**.

- As a **model** nature gives strategies that can be adopted or adapted to solve some of the problems facing man, innovative solutions should imitate or be inspired by nature's designs, processes, and ecosystems, this will ensure sustainable innovations.
- As a **measure**, nature's principles can be used as a benchmark to judge the sustainability of any innovation, and
- as a **mentor**, nature is seen as having billions of years of experience to be learnt from.

One of the early examples of biomimicry is Leonardo
Da Vinci's flying machine



Many buildings have been inspired by nature, the Eastgate building in Harare, Zimbabwe is a self-cooled high-rise building inspired by the anthill and uses 35 percent less energy than similar buildings in Zimbabwe



From anthill to the Eastgate building



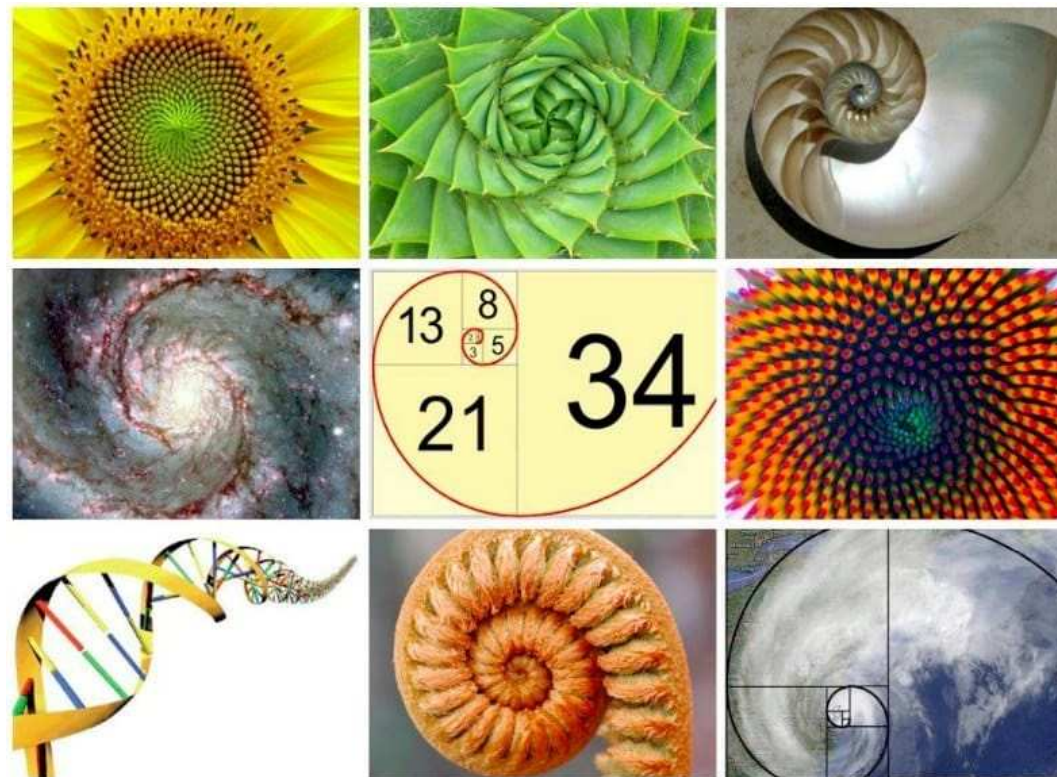
From plant with burs to Velcro

Mercedes-Benz's bionic car (4.24 m long, 4- occupant plus luggage car), inspired by the box fish (insert), had a high efficiency with 20% lower fuel consumption and 80% lower NOx emissions (when compared to cars of the same category)



Japan's 700-Series Shinkansen (the Bullet train), a fast moving train solved the 'sonic boom' problem at the exit of a tunnel by inspiration from a bird with a bullet like beak that dives swiftly and noiselessly into the water for its prey, the kingfisher. The redesigned bullet trains were 10% faster, 30% more energy efficient, and under the 70-decibel noise limit.

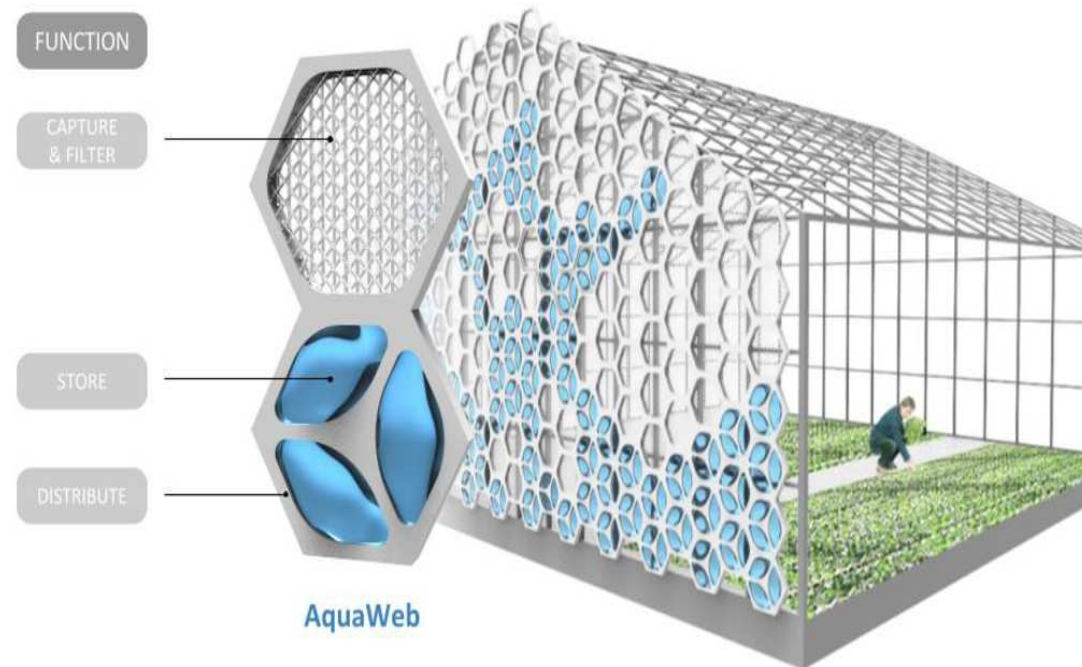
One of the most effective and common method used to solve process optimization problems is the golden section method, a line search method for one dimensional, unimodal, and non-differentiable objective functions. This method was inspired by nature's golden ratio, the Fibonacci sequence, believed to be the underlying geometry of spiral things ubiquitous in nature



This impeller, used in water storage tanks of municipal water and wastewater treatment plants, termed *the lily impeller* (right photo) was inspired by nature's ubiquitous spiral shape (slide 18) and the calla lily (left photo). The lily impeller with a height of 8 inches circulates hundreds of millions of gallons of water with the same energy footprint as a light bulb, thus reducing energy usage by 80%, noise reduction by 75%, and reducing the use of disinfectants, to treat stagnant water, by 85%.



The Aquaweb captures water from the air, stores and distributes the water saving on water and energy cost



Nature's inspiration for the AquaWeb – Spider web captures water and directs to centre, ice plant surface stores water, fungal network distributes nutrients to plants and the strong hexagonal shape of the beehive.

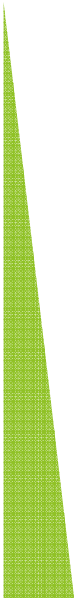
Conclusion

Remedy for saving planet earth is to emulate nature's survival strategies

The scientist, needs to study nature and have a databank of nature's time-tested patterns and strategies for the engineer to draw inspiration from.

The engineering curriculum of our universities must include nature-inspired-engineering design courses.

Nature inspired innovation, is indeed the only option left to save planet earth





Lotus leaf with water droplets and a self-cleaning material.





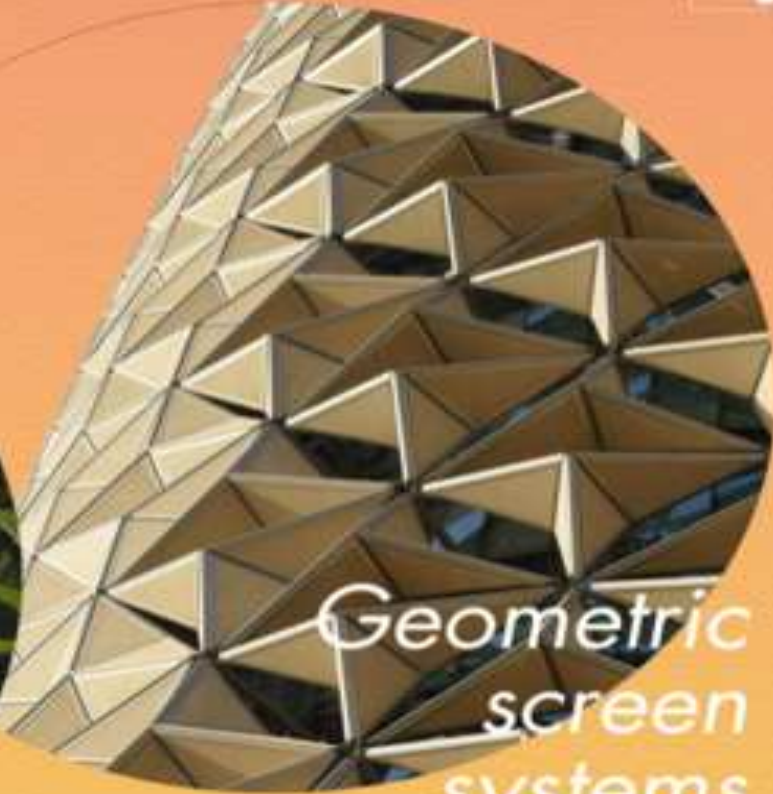
Flippers of
humpback whales

Wind
turbines

Blades of wind turbines inspired by the flippers of humpback whales



Flowers



Geometric
screen
systems

Open-close geometry of the facade inspired by the sunflower

This building screen system opens and closes in response to sunlight (like a flower). The geometry and elasticity of the flowers played a major role in the study.

Termite den



Geometric
screen systems



Temperature regulation of East Gate Office inspired by termite dens

In order to create a building without air conditioning, the East Gate Office in Zimbabwe adapted a natural cooling system inspired by dens.



Geckos feet

Climbing
pads

Adhesiveness of climbing pads inspired by the feet of Gecko

Geckos climb vertically with the help of very tiny hair-like strands on the bottom of their feet called setae. Due to weak attraction between molecules, a sticking force is created between the setae and the walls. Climbing pads, mimicking this method, capable of supporting human weight are now in the making.

Stenocara beetle



Water collector



Water absorptive surface of water collector inspired by the stenocara beetle

Researchers are trying to make synthetic surfaces inspired for the Stenocara beetle's body which helps it to capture water from air. This is done with the help of grooves on the beetle's forewings.

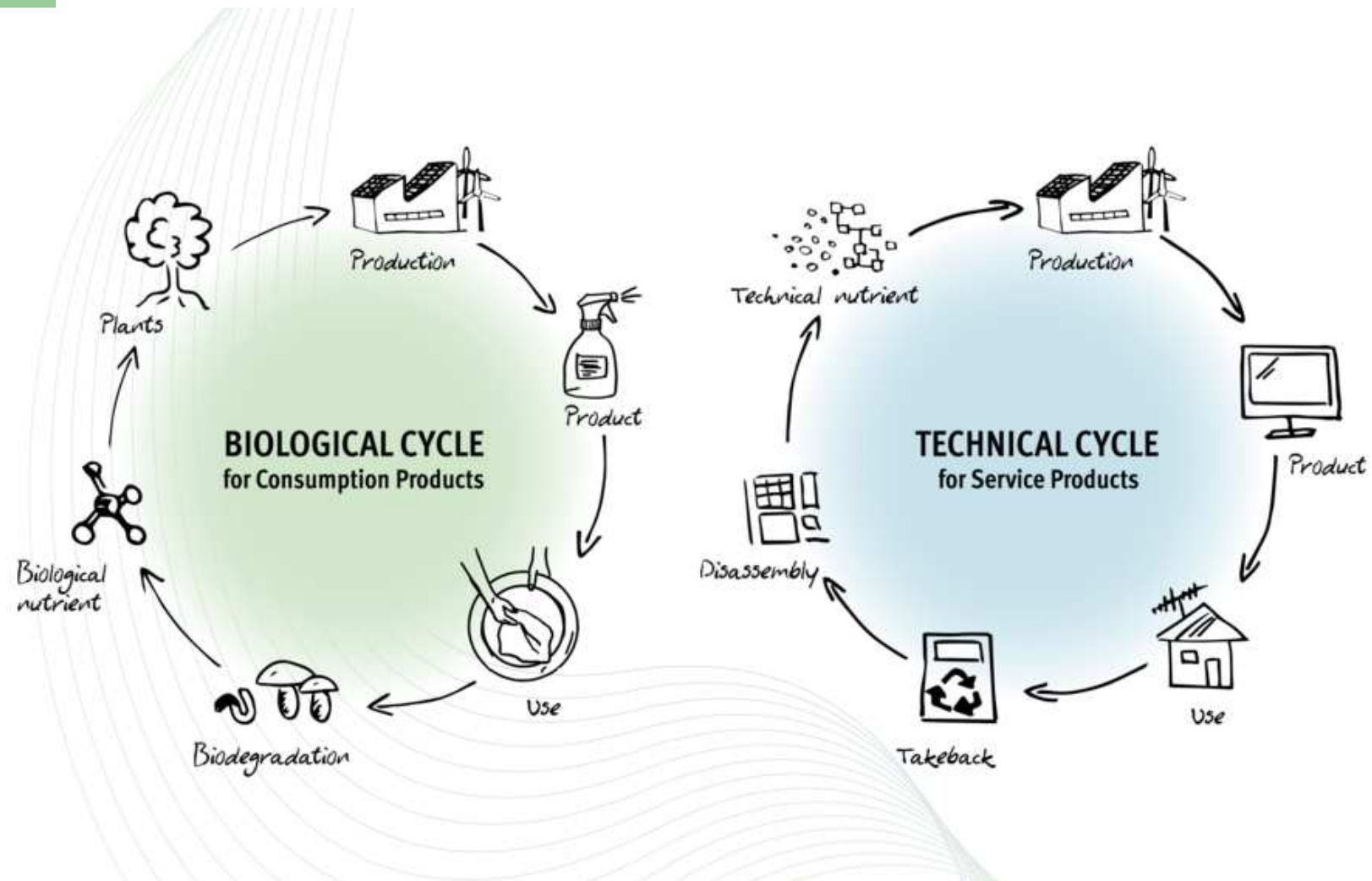






Shark skin close up and antimicrobial surface

Cradle-to-Cradle, aka C2C, or circular economy



*Consumables like natural fibres, cleaning agents or biodegradable packaging circulate in a **biological cycle** to which they can be safely reintroduced after use. They turn into compost or other materials which are in turn used to make new products. In this way, old products do not turn into waste but into “nutrients” for a new product.*



*Consumer goods such as electronic items or flooring circulate in a technical cycle. These products are already optimised during the design and manufacturing process as material resources for their next service life as new products. Components can be sorted according to their constituent materials after use and then reintroduced to a **technical cycle**. In so doing, a high materials quality is maintained and a downcycling can be prevented.*

Conclusion

Towards the end of his life, Steve Jobs said “I think the biggest innovations of the 21st century will be at the intersection of biology and technology. A new era is beginning.” This collaboration between biology and technology, in which technological innovations draw inspirations from biological and natural processes, to find solutions to some of man’s problems, could be the most effective route to sustainable development

Albert Einstein said, “*We cannot solve the problems we face in the same mindset that created them in the first place. Doing things the same way and expecting a different result is insanity*”.

